

Final Project Schedule

A Week-by-Week Guide from Question to Conclusion

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This document provides a detailed timeline for your final project. The goal is to break the project into manageable steps and ensure you are making steady progress. Following this schedule will help you produce a more thoughtful and rigorous final product.

Summary of Major Deadlines

Deliverable	Due Date	Weight
Project Proposal	End of Week 9	10%
Final Presentation	In Class, Week 14	10%
Final Paper	End of Finals Week	25%

Using Canvas for Project Deliverables

We will use Canvas to facilitate feedback and track progress throughout the semester. In general, there are two types of deliverables:

- **Discussion Boards (Peer Feedback):** For early-stage brainstorming and feedback (e.g., sharing your DAG), you will post your work to a weekly discussion board. The goal is to learn from each other and refine your ideas. Participation requires posting your own work and providing constructive feedback to at least one classmate.
- **Assignments (Check-ins & Graded Submissions):** For formal submissions, you will upload a document to a Canvas Assignment. Some of these will be ungraded check-ins to ensure you are on track (e.g., submitting a diagnostic plot), while the three major deadlines will be graded.

Part I: Foundations (Weeks 1-4)

The focus of this part is on defining a strong, feasible causal question and building the theoretical foundation for your analysis.

Week 1: The Causal Question

- **Project Focus:** Brainstorming and selecting a topic.
- **Connection to Course:** We are defining what a causal effect is. Your task is to find one you want to estimate.
- **Suggested Tasks:**
 - Explore the [Data Resources](#) page and other sources.
 - Do a quick literature search to see if your potential questions are novel and feasible.
- **Deliverable:** Post 2-3 potential causal questions to the “**Week 1: Project Brainstorming**” discussion board on Canvas.

Week 2: The DAG

- **Project Focus:** Specifying your causal assumptions.
- **Connection to Course:** We are learning the language of DAGs. You will apply it directly to your research question.
- **Suggested Tasks:**
 - Settle on a single causal question from your Week 1 list.
 - Draw a preliminary DAG including your exposure, outcome, and key potential confounders.
 - Write a short paragraph justifying the nodes and arrows.
- **Deliverable:** Post an image of your DAG and your justification paragraph to the “**Week 2: DAG Peer Feedback**” discussion board. Provide thoughtful feedback on at least one classmate’s DAG.

Week 3: Assumptions

- **Project Focus:** Articulating your identification strategy.
- **Connection to Course:** We’re formalizing exchangeability, positivity, and consistency.
- **Suggested Tasks:**
 - Using your DAG, identify a sufficient adjustment set.
 - Consider if these variables are available in your data. Note any critical unmeasured confounders.
- **Deliverable:** In a text entry, submit your identified adjustment set and a 1-2 sentence statement on the plausibility of positivity to the “**Week 3: Assumptions Check-in**” assignment (ungraded).

Week 4: Study Design

- **Project Focus:** Emulating the Target Trial.
- **Connection to Course:** We are discussing sources of bias. The target trial framework helps us proactively minimize them.
- **Suggested Tasks:**
 - Begin drafting the protocol for your hypothetical target trial.
 - Focus on Eligibility Criteria, Treatment Strategies, and Assignment Procedure.
- **Deliverable:** Post a 1-paragraph summary of your target trial's eligibility criteria to the “**Week 4: Target Trial Design**” discussion board.

Part II: Estimation (Weeks 5-9)

The focus now shifts from theory to practice. You will begin implementing methods to estimate your causal effect.

Week 5: Propensity Scores

- **Project Focus:** Modeling the exposure.
- **Connection to Course:** We are learning about propensity scores (PS). You will now estimate them.
- **Suggested Tasks:**
 - Write the code to fit your PS model.
 - Generate diagnostic plots (e.g., histograms or density plots) to check for overlap.
- **Deliverable:** Submit a PDF of your PS overlap plot to the “**Week 5: PS Overlap Check-in**” assignment (ungraded).

Weeks 6 & 7: Weighting & G-Computation

- **Project Focus:** Estimating the causal effect.
- **Connection to Course:** We are covering IPTW and G-computation.
- **Suggested Tasks:**
 - Calculate inverse probability of treatment weights (IPTW).
 - Check for covariate balance after weighting using a Love plot from the `cobalt` package.
 - Use the weights to get a preliminary estimate of the average treatment effect.
- **Deliverable:** Submit a PDF of your covariate balance (Love) plot to the “**Week 7: Covariate Balance Check-in**” assignment (ungraded).

Week 8: Instrumental Variables

- **Project Focus:** Project proposal refinement.
- **Connection to Course:** Understanding IV helps you think critically about confounding.
- **Suggested Tasks:**
 - Consolidate your causal question, DAG, target trial protocol, and proposed methods into a single document.
 - Draft all sections of the proposal.
- **Deliverable:** This is a dedicated work week. No deliverable is due as you prepare the proposal.

Week 9: Effect Modification

- **Project Focus:** Proposal Submission.
- **Connection to Course:** Thinking about heterogeneity may inform your project's secondary aims.
- **Suggested Tasks:**
 - Finalize and polish your proposal document.
- **Deliverable:** Submit your final Project Proposal PDF to the **"Project Proposal"** assignment on Canvas (**Graded**).

Part III: Refinement & Reporting (Weeks 10-14)

The final part is about ensuring your results are robust, interpreting them correctly, and communicating them effectively.

Weeks 10 & 11: Mediation

- **Project Focus:** Writing the Methods and Results sections.
- **Connection to Course:** Thinking about mediation helps refine your causal pathways.
- **Suggested Tasks:**
 - Write a full draft of your paper's **Methods** and **Results** sections.
 - Create your "Table 1" and all figures.
- **Deliverable:** This is a dedicated work week for writing. No deliverable is due.

Week 12: Sensitivity Analysis

- **Project Focus:** Assessing robustness.
- **Connection to Course:** Direct application of sensitivity analysis techniques.
- **Suggested Tasks:**
 - Calculate an E-value for your primary finding.
 - Write up the sensitivity analysis portion of your paper.
- **Deliverable:** Post your main effect estimate and its corresponding E-value, along with a 1-sentence interpretation, to the “**Week 12: E-value Check-in**” discussion board.

Week 13: Target Trial Emulation

- **Project Focus:** Finalizing the paper and presentation.
- **Connection to Course:** A synthesis of the target trial framework.
- **Suggested Tasks:**
 - Write the **Introduction** and **Discussion** sections of your paper.
 - Create the slides for your final presentation and practice your timing.
- **Deliverable:** Submit your presentation slides (PDF or PPTX) to the “**Week 13: Presentation Slides**” assignment (ungraded) so they are ready for next week.

Week 14: Final Presentations

- **Project Focus:** Communicating your research.
- **Deliverable:** In-class project presentations (**Graded**).

Finals Week

- **Project Focus:** Final revisions.
- **Suggested Tasks:**
 - Incorporate feedback from your presentation.
 - Proofread and format your paper carefully.
- **Deliverable:** Submit your Final Paper PDF to the “**Final Paper**” assignment on Canvas (**Graded**).